SVKM's Narsee Monjee Institute of Management Studies
Mukesh Patel School of Technology Management & Engineering

Program: B Tech / MBA Tech (All Program)				**Semester:** I/II	
Course/Module: Linear Algebra and Differential Equations				**Module Code:**	
Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks- 100 in Question Paper)**
3	2	0	4	Marks Scaled to 50	Marks Scaled to 50

Pre-requisite:
Knowledge of fundamental concepts in Algebra, Differential and Integral Calculus.

Course Objectives:
This course aims to instil in prospective engineers' knowledge of concepts and techniques in Linear Algebra and Differential Equations. It also prepares the students to deal with advanced level of Mathematics and applications that would be essential for their disciplines.

Course Outcomes:
After completion of the course, students would be able to-

1. Solve systems of linear equations using the matrix method and understand linear algebra concepts, including vector spaces, subspaces, basis, and dimension (K3).
2. Analyze linear transformations by understanding their matrices, kernels, ranges and compute Eigenvalues and Eigenvectors of matrices and apply them to solve problems related to diagonalization (K3).
3. Solve first-order ordinary differential equations using standard techniques for exact and linear equations (K3).
4. Utilize the operator method and the method variation of parameters to find solutions of higher-order linear differential equations and the Euler-Cauchy equation (K3).
5. Form partial differential equations and employ direct integration and separation of variables techniques to find solutions (K3).
6. Apply concepts from linear algebra and differential equations to model and address real-world problems in engineering, physics, and applied sciences (K3).

Detailed Syllabus:

Unit	Description	Duration
1	**Linear Equations and Vector Spaces** Rank of Matrix, System of linear equations.	4
	Vector space, Subspace of vector space, Linear span, Linear independence and dependence, Basis, Dimension.	6
2	**Linear Transformation and Eigenvalues** Linear transformation, Matrix associated with linear transformation, Composition of linear maps, Kernel and Range of a linear map, Rank- Nullity Theorem, Inverse of a linear transformation.	6
	Cayley- Hamilton Theorem, Eigenvalues, Eigenvectors, Eigenvalues of symmetric, skew- symmetric, Hermitian and Skew-Hermitian matrices, Diagonalization, Orthogonal Diagonalization of a real symmetric matrix.	6

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3	**First order Ordinary Differential Equations** Exact equations, Equations reducible to exact equations using integrating factors, Linear equations, Bernoulli equation, Applications of FODE.	07
4	**Higher order Ordinary Differential Equations** Higher order linear differential equations with constant coefficients, operator method.	6
	Variation of parameters method, Euler-Cauchy equation, Applications of HODE.	4
5	**Partial Differential Equations** Introduction, Formation of Partial Differential Equations, Classification of second order Partial Differential Equations, Solutions of Partial Differential Equations using variable separable method. Examples on the heat equation, wave equation and the Laplace equation.	06
	Total	**45**

Text Books

1. B.V. Ramana, Higher Engineering Mathematics, 1st Edition, McGraw Hill Education, 2017.
2. B.S. Grewal, Higher Engineering Mathematics, 44th Edition, Khanna Publishers, 2017.
3. D. Poole, Linear Algebra: A Modern Introduction, 3rd Edition, Brooks/Cole, 2010.

Reference Books

1. S. H. Friedberg, A. J. Insel, L. E. Spence, Linear Algebra, 5th Edition, Pearson, 2019.
2. G. Strang, Introduction to linear algebra, 5th Edition, Wellesley Cambridge Press, 2016.
3. G. F. Simmons, Differential equations with applications and historical notes, 2nd Edition McGraw- Hill Education, 2017.
4. T. Amarnath, An elementary course in partial differential equations, New Delhi Narosa Publishing House, 2003.
5. Paul Zimmermann, Computational Mathematics with SageMath, SIAM, 2019
6. Ajit Kumar et. al, Linear Algebra with Sage, Kyobobook, S. Korea, 2015. http://matrix.skku.ac.kr/2015-Album/Big-Book-LinearAlgebra-Eng-2015.pdf.

Laboratory Work

Minimum 8 to 10 programming exercises based on the syllabus.

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